

Detailed Functionality Document **For Hunter's Unity Mobile App**

Objective:

The objective of this project is to create a fast-paced, real-time multiplayer mobile game where four players defend their goal sides in a square arena using swipe-controlled paddles. Balls continuously bounce around the board, with new balls spawning every 10 seconds to increase difficulty. Players lose lives when a ball enters their goal, leading to eliminations until a final winner remains. The game incorporates ranking progression, bots to fill empty lobbies, extra life purchases, skins, and a Halo-style competitive leaderboard system.

Note: The client will provide all external links and any required paid APIs if necessary.

Introduction

This app is a **multiplayer online mobile game** built around fast-paced defense and elimination mechanics. Each match begins with four players, each on one side of a square board, controlling paddles by swiping on their mobile screens. Balls bounce around the arena, and players must prevent them from entering their goal or they lose a life.

Over time, new balls appear every 10 seconds, making the game more chaotic and challenging. Players can lose all their lives and get eliminated, transitioning the match stage into 3-player mode, then 2-player mode, and eventually a final showdown to determine the winner. Users can buy one-time extra lives per match, unlock paddle skins, and climb global rankings based on wins and losses.

The app provides a clean user journey: open the app → matchmaking → gameplay → win/lose outcome → ranking update → rewards. Every part of the system is designed to be smooth, beginner-friendly, and competitive. The goal is to give players a fun air-hockey-style experience that they can enjoy alone or with friends.

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Functional Requirements:

1. Core Gameplay Mechanics

1.1 Arena & Paddle System

- The arena is a **square board** with four sides, each acting as a player's goal.
- Every player has a paddle that moves **left-right** along their assigned side.
- Movement is controlled through **swipe gestures** on the screen.
- Paddle collisions are handled by a **real-time physics engine**.
- System ensures smooth, responsive paddle movement.

Integration:

Connects with ball physics, player input, lives system, and elimination logic.

1.2 Ball Physics & Spawning

- Game starts with **one ball** in the arena.
- A new ball spawns every **10 seconds**.
- Balls bounce off walls and paddles following realistic physics.
- If a ball enters a player's goal → that player loses a life.

Integration:

Works with physics engine, scoring logic, UI display, and elimination flow.

1.3 Lives System & Elimination

- Each player begins with **5 lives**.
- When a ball enters their goal, they lose **1 life**.
- Players may buy **+3 extra lives** only **once per match**.
- When lives reach 0 → the player is **eliminated**.

Integration:

Controls match stage transitions and ranking calculation.

1.4 Match Flow & Stage Transitions

- Match begins with **4 players (1v1v1v1)**.
- After one player dies → **3-player mode (1v1v1)**.
- After next elimination → **final duel (1v1)**.
- Last remaining player is the **winner**.

Integration:

Triggers ranking updates, reward distribution, and end-game animations.

2. Multiplayer & Matchmaking

2.1 Real-Time Online Multiplayer

- Uses live data synchronization.
- Ensures all paddle positions, ball movements, and collisions update instantly.
- Designed for fast-paced competitive gameplay.

2.2 Bot System

- If fewer than 4 players enter the lobby, bots fill remaining slots.
- Bots behave realistically and defend their goals.
- Difficulty can scale based on player rank or match type.

2.3 Matchmaking System

- Players join ranked or casual queues.
- Automatically sorts players into lobbies.
- Uses ranking level to create balanced matches.
- Starts match automatically once lobby is full (players + bots).

3. Ranking & Leaderboard System

3.1 Ranking Levels (1–50)

- Every player starts at **Level 1**.
- Winning increases rank, losing decreases rank.
- Ranking based on the Halo 3 style system:
 - Beat higher-level players → jump more levels.
 - Lose against lower-level players → drop more levels.

3.2 Match-Based Promotion Logic

- To move from Level 1 → Level 2:
 - You must win the **1v1v1v1**, then **1v1v1**, then **1v1**.
- If you do not win the final match, **you do not level up**.
- Losing matches at higher levels can drop a player down levels.

3.3 Leaderboards

- Global leaderboard.
- Regional leaderboard.
- Friends leaderboard.
- Displays rank, level, win rate, and profile info.

4. In-App Purchases (IAP)

4.1 Extra Lives

- Players may buy **3 additional lives**.
- Can only be bought **once per match**.
- Designed to avoid pay-to-win advantages.

4.2 Cosmetic Skins

- Paddle skins.
- Arena color themes.
- Unlockable or purchasable (no gameplay benefit).

4.3 In-App Store

- Displays purchasable items.
- Shows skins, themes, and extra lives.
- Supports Apple/Google payment systems.

5. Audio & Visual Features

5.1 Background Music

- Plays during matches and menus.
- Looping soundtrack.
- User can mute or adjust volume.

5.2 Color-Changing Board

- Arena colors shift gradually during gameplay.
- Colors may intensify with more balls or fewer players.

5.3 Sound Effects

- Paddle hits.
- Ball collisions.
- Goal/life-loss sounds.
- Match start & match end sounds.

6. User Interface & Navigation

6.1 Main Menu

Includes:

- Play Ranked
- Play Casual
- Store
- Settings
- Leaderboard
- Profile

6.2 In-Game Interface

Shows:

- Lives remaining
- Number of active balls
- Player icons and positions
- Pause button
- Match timer (optional)

6.3 Settings Menu

- Music on/off
- Sound effects on/off
- Sensitivity controls
- Account and login options

7. Account & Data Management

7.1 User Accounts

- Login via Email, Apple, Google, or Guest account.
- Saves progress and rank.

7.2 Statistics Tracking

Tracks:

- Wins and losses
- Highest rank reached
- Total games played
- Balls deflected
- Match history

7.3 Cloud Storage

- Syncs progress across devices.
- Prevents data loss.

Conclusion

In conclusion, this game concept delivers a simple yet intense real-time multiplayer experience where four players battle to defend their goals in a fast-paced, air-hockey-style arena. The increasing challenge of multiple balls, strategic life management, and dynamic eliminations keeps gameplay exciting and competitive. With a Halo-inspired ranking system, bots for full lobbies, cosmetic customization, and smooth online matchmaking, the game offers long-term engagement for both casual and competitive players. This foundation provides everything needed to build a fun, scalable, and highly replayable mobile game.